

Testing & TDD

Modul: M324 – DevOps-Prozesse mit Tools unterstützen

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Unit-Tests für bestehenden Code

Als erstes wurden Unit-Tests für den bereits bestehenden Code geschrieben. Ziel war es, das bestehende Verhalten des Backends zu prüfen.

Test 1 – Anwendung startet korrekt

Prüft ob der Spring-Kontext fehlerfrei geladen wird.

```
@Test
void contextLoads() {
    assertTrue(true);
}
```

Test 2 – Task Getter/Setter

Prüft ob die `Task`-Klasse den Wert korrekt speichert und zurückgibt.

```
@Test
void testTaskDescription() {
    Task task = new Task();
    task.setTaskdescription("Einkaufen");
    assertEquals("Einkaufen", task.getTaskdescription());
}
```

Test 3 – GET-Endpoint mit MockMvc

Prüft ob der REST-Endpoint `GET /` mit HTTP 200 antwortet.

```
@Test
void testGetEndpointReturnsOk() throws Exception {
    mockMvc.perform(get("/"))
        .andExpect(status().isOk());
}
```

```
13 @SpringBootTest
14 @AutoConfigureMockMvc
15 class DemoApplicationTests {
16
17     @Autowired
18     private MockMvc mockMvc;
19
20     // Test 1: Anwendung startet korrekt
21     @Test
22     void contextLoads() {
23         assertTrue(true);
24     }
25
26     // Test 2: Task Getter/Setter funktioniert
27     @Test
28     void testTaskDescription() {
29         Task task = new Task();
30         task.setTaskdescription(taskdescription: "Einkaufen");
31         assertEquals("Einkaufen", task.getTaskdescription());
32     }
33
34     // Test 3: GET "/" gibt HTTP 200 zurück (MockMvc)
35     @Test
36     void testGetEndpointReturnsOk() throws Exception {
37         mockMvc.perform(get(uriTemplate: "/"))
38             .andExpect(status().isOk());
39     }
40
41     // TDD Test 4: Task hat Erfassungsdatum bei Instanziierung
42     @Test
43     void testTaskHasCreationDate() {
44         Task task = new Task();
45         assertNotNull(task.getCreatedAt(), "Task soll ein Erfassungsdatum haben");
46     }
47
48 }
```

TDD – Test Driven Development

Schritt 1: Failing Test schreiben

Zuerst wurde ein Test geschrieben für eine Funktionalität, die noch nicht existiert: das automatische Speichern des Erfassungsdatums beim Erstellen eines Tasks.

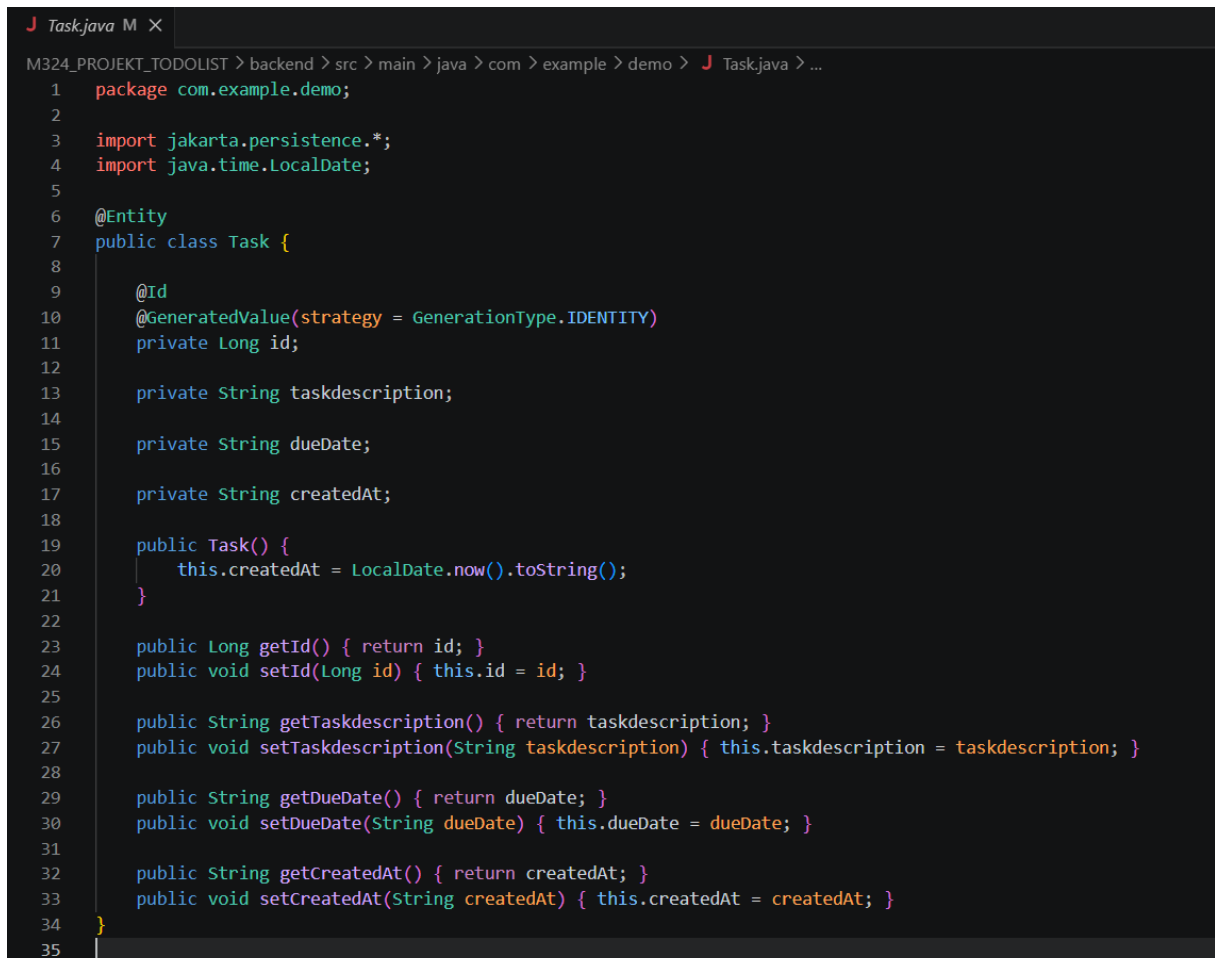
```
@Test
void testTaskHasCreationDate() {
    Task task = new Task();
    assertNotNull(task.getCreatedAt(), "Task soll ein Erfassungsdatum haben");
}
```

Dieser Test schlägt fehl, da `getCreatedAt()` in der `Task`-Klasse noch nicht existiert.

Schritt 2: Funktionalität implementieren

Die `Task`-Klasse wurde erweitert: ein `createdAt`-Feld wird beim Instanziiieren automatisch mit dem aktuellen Datum befüllt.

```
private String createdAt;  
  
public Task() {  
    this.createdAt = LocalDate.now().toString();  
}  
  
public String getCreatedAt() { return createdAt; }  
public void setCreatedAt(String createdAt) { this.createdAt = createdAt; }
```



```
J Task.java M X  
M324_PROJEKT_TODO LIST > backend > src > main > java > com > example > demo > J Task.java > ...  
1 package com.example.demo;  
2  
3 import jakarta.persistence.*;  
4 import java.time.LocalDate;  
5  
6 @Entity  
7 public class Task {  
8  
9     @Id  
10    @GeneratedValue(strategy = GenerationType.IDENTITY)  
11    private Long id;  
12  
13    private String taskdescription;  
14  
15    private String dueDate;  
16  
17    private String createdAt;  
18  
19    public Task() {  
20        this.createdAt = LocalDate.now().toString();  
21    }  
22  
23    public Long getId() { return id; }  
24    public void setId(Long id) { this.id = id; }  
25  
26    public String getTaskdescription() { return taskdescription; }  
27    public void setTaskdescription(String taskdescription) { this.taskdescription = taskdescription; }  
28  
29    public String getDueDate() { return dueDate; }  
30    public void setDueDate(String dueDate) { this.dueDate = dueDate; }  
31  
32    public String getCreatedAt() { return createdAt; }  
33    public void setCreatedAt(String createdAt) { this.createdAt = createdAt; }  
34 }  
35
```

Schritt 3:

Alle Tests bestehen

Nach der Implementierung wurden alle 4 Tests mit einem einzigen Aufruf ausgeführt:

```
mvn test
```

```
PS C:\Users\Metehan Celik\Desktop\M324-ToDo\M324_PROJEKT_TODOLIST\backend> mvn test
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 5.439 s -- in com.example.demo.DemoApplicationTests
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO]
[INFO] Total time: 7.345 s
[INFO] Finished at: 2026-05-22T17:39:49+02:00
[INFO]
[INFO] -----
PS C:\Users\Metehan Celik\Desktop\M324-ToDo\M324_PROJEKT_TODOLIST\backend>
```

Applikation startet fehlerfrei

Nach allen Anpassungen wurde die Applikation gestartet, um zu prüfen ob keine Spring-Boot- oder CodingProbleme bestehen:

```
mvn spring-boot:run
```

```
PS C:\Users\Metehan Celik\Desktop\M324-ToDo\M324_PROJEKT_TODOLIST\backend> mvn spring-boot:run
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:demo >-----
[INFO] Building demo 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] >>> spring-boot:3.4.5:run (default-cli) > test-compile @ demo >>>
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ demo ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ demo ---
[INFO] Copying 1 resource from src\test\resources to target\test-classes
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] <<< spring-boot:3.4.5:run (default-cli) < test-compile @ demo <<<
[INFO]
[INFO]
[INFO] --- spring-boot:3.4.5:run (default-cli) @ demo ---
[INFO] Attaching agents: []

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:: Spring Boot ::                (v3.4.5)
```

```

PS C:\Users\Metehan Celik\Desktop\M324-Todo\M324-PROJEKT_TODOLIST\backend> mvn spring-boot:run
2026-05-22T17:40:50.784+02:00 INFO 24308 --- [main] com.example.demo.DemoApplication : Starting DemoApplication using Java 21.0.7 with PID 24308 (C:\Users\Metehan Celik\Desktop\M324-Todo\M324-PROJEKT_TODOLIST\backend)
2026-05-22T17:40:50.786+02:00 INFO 24308 --- [main] com.example.demo.DemoApplication : started by Metehan Celik in C:\Users\Metehan Celik\Desktop\M324-Todo\M324-PROJEKT_TODOLIST\backend
2026-05-22T17:40:51.253+02:00 INFO 24308 --- [main] .s.d.r.c.RepositoryConfigurationDelegate : No active profile set, falling back to 1 default profile: "default"
2026-05-22T17:40:51.305+02:00 INFO 24308 --- [main] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-05-22T17:40:51.639+02:00 INFO 24308 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Finished Spring data repository scanning in 44 ms. Found 1 JPA repository interface.
2026-05-22T17:40:51.648+02:00 INFO 24308 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-05-22T17:40:51.648+02:00 INFO 24308 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-05-22T17:40:51.648+02:00 INFO 24308 --- [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.40]
2026-05-22T17:40:51.687+02:00 INFO 24308 --- [main] o.a.c.c.c.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-05-22T17:40:51.689+02:00 INFO 24308 --- [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 867 ms
2026-05-22T17:40:51.800+02:00 INFO 24308 --- [main] o.hibernate.jpa.internal.util.LogHelper : HHH000204: Processing PersistenceUnitInfo [name: default]
2026-05-22T17:40:51.832+02:00 INFO 24308 --- [main] org.hibernate.Version : HHH000012: Hibernate ORM core version 6.6.13.Final
2026-05-22T17:40:51.857+02:00 INFO 24308 --- [main] o.h.c.internal.RegionFactoryInitiator : HHH000026: Second-level cache disabled
2026-05-22T17:40:52.058+02:00 INFO 24308 --- [main] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class transformer
2026-05-22T17:40:52.082+02:00 INFO 24308 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-05-22T17:40:52.228+02:00 INFO 24308 --- [main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection conn0: url=jdbc:h2:file:./data/tododb user=SA
2026-05-22T17:40:52.229+02:00 INFO 24308 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-05-22T17:40:52.244+02:00 WARN 24308 --- [main] org.hibernate.orm.deprecation : HHH0000025: H2Dialect does not need to be specified explicitly using 'hibernate.dialect'
(remove the property setting and it will be selected by default)
2026-05-22T17:40:52.262+02:00 INFO 24308 --- [main] org.hibernate.orm.connections.pooling : HHH10001005: Database info:
Database JDBC URL [connecting through datasource 'HikariDataSource (HikariPool-1)']
Database driver: undefined/unknown
Database version: 2.3.232
Autocommit mode: undefined/unknown
Isolation level: undefined/unknown
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2026-05-22T17:40:52.760+02:00 INFO 24308 --- [main] o.h.e.t.j.p.i.JtaPlatformInitiator : HHH000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA platform integration)
2026-05-22T17:40:52.806+02:00 INFO 24308 --- [main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-05-22T17:40:53.002+02:00 WARN 24308 --- [main] jpaBaseConfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default. Therefore, database queries may be performed during view rendering. Explicitly configure spring.jpa.open-in-view to disable this warning
2026-05-22T17:40:53.384+02:00 INFO 24308 --- [main] o.s.b.a.h2.H2ConsoleAutoConfiguration : H2 console available at '/h2-console'. Database available at 'jdbc:h2:file:./data/tododb'
2026-05-22T17:40:53.432+02:00 INFO 24308 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-05-22T17:40:53.437+02:00 INFO 24308 --- [main] com.example.demo.DemoApplication : Started DemoApplication in 2.929 seconds (process running for 3.137)

```

Test-Konfiguration

Für die Tests wird eine separate `application.properties` unter `src/test/resources/` verwendet, damit die Tests mit einer In-Memory H2-Datenbank laufen und nicht die produktive Datei beeinflussen:

```

spring.datasource.url=jdbc:h2:mem:testdb
spring.jpa.hibernate.ddl-auto=create-drop

```